

Growing Hypergraphs with Node Labels

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We study a mechanistic model of growing hypergraphs in which edge formation is tuned by homophily between binary node attributes. Edges form in this model as noisy copies of previous edges, where the transmission of nodes from one edge to the next depends on the labels. We derive a power law for the degree distribution in this model and describe the long-term dynamics of the joint distribution of labels contained in edges. This model defines a likelihood over a labeled hypergraph, allowing us to use standard maximum-likelihood techniques to motivate algorithms. We estimate the model parameters on synthetic and real data via (stochastic) expectation maximization, allowing us to statistically quantify the signature of homophily in empirical polyadic systems. We also demonstrate an approach to community detection via simulated annealing which, though computationally expensive, achieves competitive results on certain data sets known to be challenging to standard community detection techniques.

I. INTRODUCTION

Hypergraphs provide a natural, flexible data representation for complex systems with polyadic (multi-way) interactions, such as social interactions (both in-person [1] and online [2]), academic co-authorship [3], and ecological systems [4]. Much recent work has emphasized the importance of faithfully representing polyadic structure rather than reducing it to dyadic interactions [5, 6], though recent work has shown that some purportedly higher-order dynamics can be faithfully represented on dyadic networks under appropriate models [7].

Much attention in the theory of (hyper)graph mining has focused on developing novel techniques for community detection (also often called *clustering* or *partitioning*). Many but not all of these methods can be viewed as exact or approximate inference algorithms for stochastic generative models. A stochastic generative model is a parametrized probability distribution defined over hypergraphs. In one standard framework, a set $\mathcal{V}$ of n nodes is initialized along with a vector $\mathbf{z} \in [k]^n$ that assigns to each node one of k community labels. One then forms the edge set $\mathcal{E}$ iteratively by visiting each subset $R \subseteq \mathcal{V}$ and independently assigning a_R (possibly repeated) edges to R according to a distribution $p(a_R|\mathbf{z}_R, \theta)$, where $\mathbf{z}_R$ is the vector of labels of the nodes in R and θ is a set of model parameters. This results in a probability distribution over hypergraphs which can be factorized into a product of edge probabilities conditioned on the node labels and model parameters:

$$p(\mathcal{H}|\mathbf{z}, \theta) = \prod_{R \subseteq \mathcal{V}} p(a_R|\mathbf{z}_R, \theta). \quad (1)$$

Many generative models for community detection fall into this category, including hypergraph stochastic blockmodels [8–10] and some generalizations [11, 12]. The conditional independence property eq. (1) is extremely useful for computation, as factorizability of the likelihood enables many efficiencies in various inference

algorithms, including approximate maximum-likelihood estimation via direct optimization [8] or expectation-maximization [9]; spectral methods [13–15]; and message passing [16]. Another recent optimization-based approach [17] is equivalent to a microcanonical version of the stochastic blockmodel, which satisfies an asymptotic form of edge-independence in the large sparse limit. Even a recent latent-space model [18] treats edges as independent samples, albeit from a very differently stochastic generating process.

Edge independence, however, is a strong and often unrealistic assumption for modeling hypergraph data. A parallel line of work has found that polyadic data sets often contain significant edge correlations [2, 19, 20], typically in the form of *similarity between interactions*. When an interaction occurs on a set $R \subseteq \mathcal{V}$, this increases the likelihood of observing a new interaction on a set $S \subseteq \mathcal{V}$ correlated with R in the sense that $R \cap S$ is larger than would be expected by chance. These overlaps can be shown to effect the kinds of dynamics which can unfold on the hypergraph [21, 22]. Several generative models for the growth of hypergraphs with edge-dependence have been proposed [23–27]. Several of these models express dependence through a copying or recombination mechanism, in which nodes collected from one or more existing hyperedges are gathered and noisily copied into a new edge as a block.

In this paper, we introduce a model of binary node-labeled hypergraphs in which edges form via a noisy copying interaction which depends the labels. Our work is organized as follows. In Section II, we describe our model and derive limiting descriptions of the joint distribution of node labels within a given edge, as well as the degree distribution of the hypergraph. Evaluating the likelihood of our model requires marginalizing over a set of latent variables governing the noisy copy mechanism, which is computationally expensive. To estimate our model’s parameters when node labels are known, we therefore describe a stochastic expectation-maximization algorithm in which each iteration has sublinear cost in the number

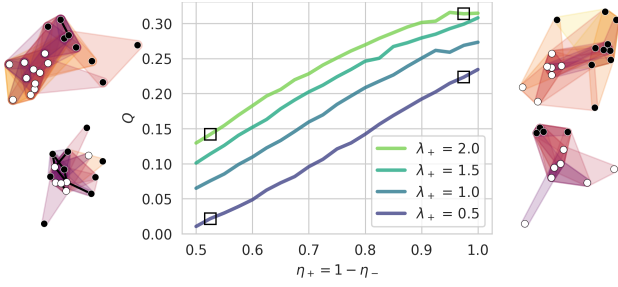


FIG. 1. Suite of model simulations illustrating the dependence of community structure, as measured by clique-projection modularity Q , on the model's parameters. The parameters $\gamma_- = 0.5$, $\eta_+ = 0.2$, and $\eta_- = 0.1$ are held fixed. Each line corresponds to a different value of γ_+ indicated by the legend. Along the horizontal axis, we vary ρ_+ and set $\rho_- = 1 - \rho_+$. Each parameter combination is simulated 10^2 times for 10^3 timesteps. On either side of the plot we show small hypergraphs sampled from parameter combinations indicated by the square markers. **PC: Fig is good but maybe we want to do 10^3 reps just to reduce the noise a bit, and possibly increase the resolution of the blue parameters.**

of edges for sparse hypergraphs. We also describe a community detection algorithm for estimating node labels via simulated annealing. In Section IV we demonstrate our algorithms on synthetic and empirical data. We show that, on synthetic data sampled from our model, our stochastic expectation maximization algorithm is both consistent and scalable, and that our simulated annealing algorithm for community detection, though computationally expensive, outperforms other standard methods. On empirical data, our parameter estimates yield insight into the mechanisms governing edge formation. Our community detection achieves competitive recovery of observed metadata labels on some data sets known to be challenging to other community detection methods. We conclude with some discussion of future directions for models of node-labeled hypergraphs with edge-dependence.

II. MODEL DESCRIPTION

Our model builds on the Hyperedge Copy Model (HCM) of He et al. [25]. In their model, edges form through a combination of three distinct steps: a copy step, in which a previous edge is sampled and imperfectly copied; an extant node addition step, in which new nodes are added to the new edge from the existing node set; and a new node addition step, in which new nodes are added to the new edge from outside the existing node set. Our model replicates these mechanisms but adds label-aware parameters to regulate each.

Formally, a state of our model at time t is a hypergraph $\mathcal{H}^{(t)} = (\mathcal{V}^{(t)}, \mathcal{E}^{(t)})$, where $\mathcal{V}$ is the set of nodes and $\mathcal{E}$ is the set of hyperedges. Each node $v \in \mathcal{V}$ has a binary label $z_v \in \{0, 1\}$. We let $\bar{z} = 1 - z$ denote the opposite label

of z . At timestep t , add edge e to $\mathcal{E}^{(t)}$ to obtain $\mathcal{E}^{(t+1)}$, where e is formed according to the following three-step process:

- **Edge-Copying:** An edge f is sampled uniformly at random from $\mathcal{E}^{(t-1)}$, and a node u is sampled uniformly at random from f . Then, each node v in f is included in e with probability ρ_+ if $z_v = z_u$ and with probability ρ_- if $z_v \neq z_u$.
- **Extant Node Addition:** We sample X_{z_u} nodes from $\mathcal{V} \setminus f$ with label z_u and add them to e , where X_z is a Poisson random variable with mean γ_+ . We similarly sample $X_{\bar{z}_u}$ nodes from $\mathcal{V} \setminus f$ with label $\bar{z}_u$ and add them to e , where $X_{\bar{z}_u}$ is a Poisson random variable with mean γ_- .
- **Novel Node Addition:** We sample Y_{z_u} new nodes with label z_u and add them to e , where Y_z is a Poisson random variable with mean η_+ . We similarly sample $Y_{\bar{z}_u}$ new nodes with label $\bar{z}_u$ and add them to e , where $Y_{\bar{z}_u}$ is a Poisson random variable with mean η_- .

In our simulations we initialize $\mathcal{H}^{(0)}$ with a single edge containing one node of each label, but any combination of hypergraph and node labels can be used as an initial condition. Both our simulations and our statistical inference techniques (described below) were carried using the XGI package for polyadic data analysis in Python [28].

Figure 1 illustrates role of the model parameters in tuning the community structure of realized hypergraphs. We measure community structure by computing Newman-Girvan modularity [29] on the clique-projection of the hypergraph. For fixed γ_- , η_+ , and η_- , increasing the difference $\rho_+ - \rho_-$ leads to higher modularity, as does increasing γ_+ . Hypergraphs sampled from models with high values of ρ_+ and γ_+ tend to have label-homogeneous cores of densely-overlapping hyperedges, which are weakly connected to one another by edges with more heterogeneous label compositions. Hypergraphs with lower values of ρ_+ and γ_+ tend to have less-structured distributions of labels over hyperedges.

A. Asymptotic Properties

1. Joint Distribution of Labels in Edges

The asymptotics of this model can be described in terms of the joint distribution $p_{K_0, K_1}(k_0, k_1)$ of nodes of labels 0 and 1 in a uniformly random edge. This joint distribution can be obtained as the leading eigenvector of a linear map. Let $p_{K_0, K_1}^{(t)}(k_0, k_1)$ be the joint distribution of label counts in a uniformly random edge after t timesteps. Then, the probability $q(k'_0, k'_1)$ of realizing an edge with k'_0 nodes of label 0 and k'_1 nodes of label 1 at

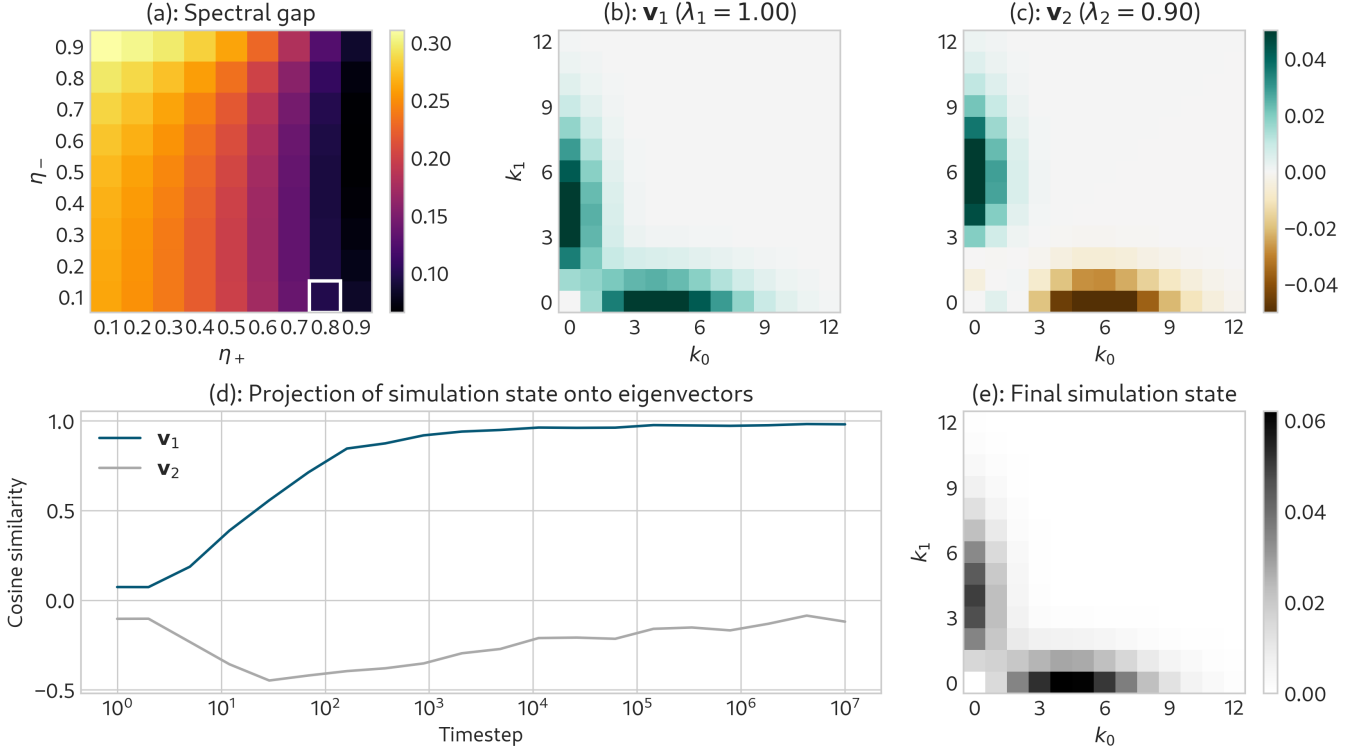


FIG. 2. Visualization of the long-run dynamics of our model, for parameters. (a): Spectral gap $1 - \lambda_2$ of the linear map T as a function of ρ_+ and ρ_- for fixed parameter values $\gamma_+ = 0.5$, $\gamma_- = \eta_- - 3$, $\eta_+ = 0.4$ and $\eta_- = 0.2$. The highlighted values $\rho_+ = 0.8$ and $\rho_- = 0.1$ are used in the remainder of the visualization. (b): Leading eigenvector $\mathbf{v}_1$ of T , describing the stationary joint distribution $p_{K_0, K_1}^{(\infty)}(k_0, k_1)$ of label counts in edges. (c): Second eigenvector $\mathbf{v}_2$ of T . The transients of the model dynamics are dominated by the subspace spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$. (d): Euclidean projections of the simulated joint distribution $p_{K_0, K_1}^{(t)}(k_0, k_1)$ onto the eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ during a simulation run with 10^7 timesteps. At each time point, we aggregate the most recent 1,000 edges to compute the joint distribution in order to approximate an “instantaneous” simulation state. (e): Empirical joint distribution $p_{K_0, K_1}^{(t)}(k_0, k_1)$ aggregated across 10^7 timesteps.

time $t + 1$ is given by

$$q(k'_0, k'_1) = \sum_{k_0, k_1} T(k'_0, k'_1 | k_0, k_1) p_{K_0, K_1}^{(t)}(k_0, k_1),$$

where $T(k'_0, k'_1 | k_0, k_1)$ gives the probability that the edge e realized in step $t + 1$ has k'_0 nodes of label 0 and k'_1 nodes of label 1 given that the edge f sampled to form e has k_0 nodes of label 0 and k_1 nodes of label 1. At stationarity, we must have $q(k'_0, k'_1) = p_{K_0, K_1}^{(t+1)}(k'_0, k'_1)$, so the stationarity condition can be ob-

tained by solving the linear system

$$p_{K_0, K_1}^{(t+1)}(k'_0, k'_1) = \sum_{k_0, k_1} T(k'_0, k'_1 | k_0, k_1) p_{K_0, K_1}^{(t)}(k_0, k_1).$$

The stationary joint distribution $p_{K_0, K_1}^{(\infty)}(k_0, k_1)$ is therefore given by the leading eigenvector of the doubly-indexed matrix T with entries $T(k'_0, k'_1 | k_0, k_1)$. The entries of this matrix can be computed analytically. Conditional on choosing an edge f with k_0 nodes of label 0 and k_1 nodes of label 1, a node of label z is selected as the seed node u with probability $\frac{k_z}{k_0 + k_1}$. Then, the number of nodes of labels z and $\bar{z}$ in e have distributions

$$\begin{aligned} k'_z &\sim 1 + \text{Binom}(k_z - 1, \rho_+) + \text{Poisson}(\gamma_+ + \eta_+), \\ k'_{\bar{z}} &\sim \text{Binom}(k_{\bar{z}}, \rho_-) + \text{Poisson}(\gamma_- + \eta_-). \end{aligned}$$

The probability of obtaining k'_0 nodes of label 0, conditioning on k_0 and k_1 , is

$$t(k'_0|k_0, k_1) = \frac{k_0}{k_0 + k_1} \sum_{i=0}^{k'_0-1} \binom{k_0-1}{i} \rho_+^i (1-\rho_+)^{k_0-1-i} e^{-(\gamma_+ + \eta_+)} \frac{(\gamma_+ + \eta_+)^{k'_0-1-i}}{(k'_0-1-i)!} +$$

$$\frac{k_1}{k_0 + k_1} \sum_{i=0}^{k'_0} \binom{k_0}{i} \rho_-^i (1-\rho_-)^{k_0-i} e^{-(\gamma_- + \eta_-)} \frac{(\gamma_- + \eta_-)^{k'_0-i}}{(k'_0-i)!}.$$

A parallel expression describes $t(k'_1|k_0, k_1)$, and the transition matrix entries are given by the product

$$T(k'_0, k'_1|k_0, k_1) = t(k'_0|k_0, k_1) t(k'_1|k_0, k_1).$$

In Figure 2, we study the dynamics of our model through the matrix T . For fixed values of the parameters γ_+ , γ_- , η_+ and η_- , the spectral gap $1 - \lambda_2$ of T is highest when ρ_- is large and ρ_+ is small. High values of the spectral gap correspond to fast convergence to the stationary distribution $p_{K_0, K_1}^{(\infty)}(k_0, k_1)$ given by the Perron eigenvector $\mathbf{v}_1$ of T with eigenvalue 1 (panel (b)). We instead illustrate an alternative regime of relatively slow convergence in which ρ_+ is large and ρ_- is small. In this regime, the simulation may maintain transient dynamics in the direction of the second eigenvector $\mathbf{v}_2$ of T with eigenvalue $\lambda_2 < 1$ (panel (c)) for relatively long periods of time before converging to the stationary distribution. We illustrate this behavior in panel (d), where the system converges relatively slowly to the leading eigenvector $\mathbf{v}_1$, resulting in a final simulation that matches it closely (panel (e)).

Figure 2 shows that, when ρ_+ is sufficiently large, the stationary distribution $p_{K_0, K_1}^{(\infty)}(k_0, k_1)$ is concentrated on edges in which typical edges have large majorities of nodes with the same label (panel b), thus displaying a form of higher-order homophily [30]. The stationary distribution is symmetric, implying that neither node label is favored in the long-run. The second eigenvector v_2 (panel c), however, is antisymmetric, describing a transient in which one a single label tends to dominate most edges. When ρ_+ is sufficiently large, the spectral gap $1 - \lambda_2$ is small (panel a), indicating that this transient decays slowly and that the system may be dominated by edges with a single majority label for relatively long periods of time before equilibrating to the leading eigenvector (panels d and e).

2. Degree Distribution

The degree distribution of a hypergraph generated from our model follows an asymptotic power law with an exponent which can be computed in closed form from the model parameters and the stationary joint distribution $p_{K_0, K_1}^{(\infty)}(k_0, k_1)$. Our derivation follows that of He et al. [25], which in turn is based on a derivation by Mitzenmacher [31]. To carry out the argument, we require a mean-field assumption that, when a node v is

selected from edge f in the edge-sampling step, the degree d of v is independent of the ratio $\frac{k_0}{k}$ of label-0 nodes in the edge f sampled to form the new edge e .

Under this assumption, it is possible to show that the degree distribution follows a power law with exponent

$$\zeta = 1 + \frac{\langle k \rangle}{1 + \rho_+ (\rho_{00} + \rho_{11} - 1) + 2\rho_- \rho_{01}} \quad (2)$$

$$\rho_{ij} = \left\langle \frac{k_i k_j}{k} \right\rangle.$$

In the expression for ρ_{ij} (and for $\langle k \rangle$) the expectation is taken with respect to the stationary joint distribution $p_{K_0, K_1}^{(\infty)}(k_0, k_1)$. Here, η can be interpreted as the expected number of novel nodes added per new edge, while σ can be interpreted as the expected number of nodes copied into a new edge from the sampled edge, normalized by the average degree of a node in the hypergraph. A derivation of this result and computational illustration are provided in Section A and Figure 6.

3. Community Structure

Show a plot describing how the modularity of the clique projection varies by parameter, and perhaps some sample hypergraphs with associated partitions.

III. MODEL INFERENCE

A. Parameter Estimation via Stochastic EM

We first consider the problem of learning the parameter vector $\boldsymbol{\theta} = (\rho_+, \rho_-, \gamma_+, \gamma_-, \eta_+, \eta_-)$ given an observed hypergraph $\mathcal{H}$ with sequential edge indices and known node labels $\mathbf{z}$. We aim to do this via maximum-likelihood estimation. Computing the complete data likelihood requires marginalizing over the model latent variables, which in this case are the edge f and node u selected at each timestep of the growth model to form the new edge e . For $e \in \mathcal{E}$, let t_e denote the time at which e was added to the hypergraph. We say that $f \prec e$ if $t_f < t_e$. Then, the complete data likelihood can be written

$$\mathbb{P}(\mathcal{H}, \mathbf{z}|\boldsymbol{\theta}) = \prod_{e \in \mathcal{E}} \sum_{f \prec e} \sum_{u \in f} p(e|f, u, \mathbf{z}, \boldsymbol{\theta}) p(f, u|\mathcal{E}_{t-1}, \mathbf{z}, \boldsymbol{\theta}). \quad (3)$$

For notational simplicity, we have repressed the additional entries of $\mathbf{z}$ introduced by the novel node addition step. The requirement to sum over the latent variables at each timestep implies a computational cost to evaluate $\mathbb{P}(\mathcal{H}, \mathbf{z}|\boldsymbol{\theta})$ (or its derivatives) which is quadratic in the number of edges, making direct optimization of the likelihood expensive for data sets of even modest size. Batch expectation-maximization [32] is an alternative approach which maximizes the likelihood without explicitly computing eq. (3), but even this approach requires forming a belief distribution over the values of all latent variables at each timestep, which again incurs quadratic cost.

We therefore use a stochastic expectation-maximization (SEM) algorithm [33] to estimate the parameters $\boldsymbol{\theta}$. Each iteration of the SEM algorithm consists of two steps: the E-step, in which exponentially-weighted moving averages of the sufficient statistics are calculated using the current parameter estimates, and the M-step, in which the parameter estimates are updated using the expected sufficient statistics.

To initialize the SEM algorithm, we initialize a vector $\mathbf{s}^{(0)}$ of sufficient statistics to arbitrary values; in our model there is a sufficient statistic for each parameter. We then update a moving average of $\mathbf{s}$ as follows. In each algorithmic step ℓ , we pick an edge e uniformly at random from $\mathcal{H}$. Then, given current estimates $\hat{\boldsymbol{\theta}}^{(\ell)}$, we form a belief distribution over the latent identities of $f \in \mathcal{E}$ and $u \in f$ which could have been selected in the edge-copying step to form e . Given e , the probability that f and u were selected can be computed in closed form. Let $m^{(t-1)}$ denote the size of the edge set at time $t-1$.

We have

$$p(f, u|\mathcal{E}^{(t-1)}, \mathbf{z}, \hat{\boldsymbol{\theta}}^{(\ell)}) = \frac{1}{|f|m^{(t-1)}} \triangleq r_{t-1}(f),$$

so Bayes' rule gives

$$p(f, u|e, \hat{\boldsymbol{\theta}}^{(\ell)}) = \frac{p(e|f, u, \mathbf{z}, \hat{\boldsymbol{\theta}}^{(\ell)})r_{t-1}(f)}{\sum_{f' \prec e} \sum_{u' \in f'} p(e|f', u', \mathbf{z}, \hat{\boldsymbol{\theta}}^{(\ell)})r_{t-1}(f')}. \quad (4)$$

A helpful note for computation is that $p(e|f, u, \mathbf{z}, \hat{\boldsymbol{\theta}}^{(\ell)}) = 0$ if either $u \notin e \cap f$ or $f \not\prec e$. We then form the vector of expected sufficient statistics

$$\mathbf{s}' = \sum_{f \prec e} \sum_{u \in f \cap e} p(f, u|e, \hat{\boldsymbol{\theta}}^{(\ell)}) \sigma(e, f, u, \mathbf{z}), \quad (5)$$

where $\sigma(e, f, u, \mathbf{z})$ is the vector of sufficient statistics for the parameters $\boldsymbol{\theta}$ given e , f , u , and $\mathbf{z}$. The entries of $\sigma(e, f, u, \mathbf{z})$ can be computed as functions of the node labels in e and f ; we supply explicit formulae in Section B. We then perform the update

$$\mathbf{s}^{(\ell+1)} = (1 - \alpha^{(\ell)})\mathbf{s}^{(\ell)} + \alpha^{(\ell)}\mathbf{s}',$$

where $\alpha^{(\ell)}$ is the learning rate at step ℓ ; we use an exponentially decaying learning rate in our experiments.

Our estimate of the parameter vector $\boldsymbol{\theta}$ is then given by a function $\hat{\boldsymbol{\theta}}^{(\ell+1)} = g(\mathbf{s}^{(\ell+1)})$ of the sufficient statistics, which we also supply in Section B. Algorithm 1 summarizes the steps of this algorithm. Our version of the algorithm uses an online computation of $\mathbf{s}'$, which allows us to perform the double-sum appearing in eqs. (4) and (5) only once, and without explicitly storing the resulting belief distribution at any point.

Algorithm 1 SEM Update Step

Require: $\mathcal{H} = (\mathcal{V}, \mathcal{E})$, $\mathbf{z}$, $\hat{\boldsymbol{\theta}}^{(\ell)}$, $\mathbf{s}^{(\ell)}$
 $e \leftarrow \text{Uniform}(\mathcal{E})$
 $\mathbf{a} \leftarrow \mathbf{0}$
 $\mathbf{b} \leftarrow \mathbf{0}$
for $f \in \mathcal{E}$ **do**
 if $f \prec e$ **then**
 for $u \in e \cap f$ **do**
 $\mathbf{a} \leftarrow \mathbf{a} + p(f, u|e, \hat{\boldsymbol{\theta}}^{(\ell)})\sigma(e, f, u, \mathbf{z})$
 $\mathbf{b} \leftarrow \mathbf{b} + p(f, u|e, \hat{\boldsymbol{\theta}}^{(\ell)})$
 $\mathbf{s}' \leftarrow \mathbf{a}/\mathbf{b}$
 $\mathbf{s}^{(\ell+1)} \leftarrow (1 - \alpha^{(\ell)})\mathbf{s}^{(\ell)} + \alpha^{(\ell)}\mathbf{s}'$
 $\hat{\boldsymbol{\theta}}^{(\ell+1)} \leftarrow g(\mathbf{s}^{(\ell+1)})$
return $\hat{\boldsymbol{\theta}}^{(\ell+1)}$, $\mathbf{s}^{(\ell+1)}$

The SEM algorithm continues iterating until the stopping criterion is met. In our experiments, the stopping criterion is met when each of the current parameter estimates $\hat{\boldsymbol{\theta}}^{(\ell)}$ is within [THRESH] of the [LENGTH] preceding estimates of the corresponding parameter.

B. Community Detection via MLE

Equation (3) defines a likelihood function over both the parameter vector $\boldsymbol{\theta}$ and the node labels $\mathbf{z}$. This allows us to attempt to also learn $\mathbf{z}$ via maximum-likelihood estimation. Unlike in stochastic blockmodel variants in which there are several heuristics with favorable computational properties [8, 9, 12, 13, 15, 17, 34], the dependence structure encoded by our model makes it difficult to apply standard efficient heuristics; the development of such heuristics would be an important direction of future work. In this study, we therefore attempt to learn $\mathbf{z}$ via simulated annealing [35], which is a general-purpose optimization algorithm that can be used to approximately maximize the likelihood over $\mathbf{z}$. Our primary aim is to demonstrate that likelihood maximization in this model can detect label structure in at least some circumstances where more standard community detections may fail, thereby motivating the development of more efficient optimization heuristics in future work.

A direct evaluation of eq. (3) requires $O(m^2)$ terms, making it expensive to evaluate even for data sets of modest size. An important computational limitation of simulated annealing in this context is that the label z_i of node i appears in *all* terms of the likelihood eq. (3) corresponding to edges after which node i arrives. This

includes edges in which node i does not participate, implying that we must perform $O(m^2)$ computations even to evaluate the change in likelihood under a proposed change to z_i .

To mitigate the computational burden, we use an intersection-based approximation to the likelihood of forming edge e in which we only sum over candidate copied edges f such that $f = \operatorname{argmax}_{f' \in \mathcal{E}} |f' \cap e|$. To justify this approximation, we start by observing that, if f is the edge sampled to form e , then the number of extant nodes required to form e is $x = |e| - |e \cap f| - |e \setminus \mathcal{V}|$. We'll argue that, when the number of nodes n is large, $p(e|f, u, \mathbf{z}, \boldsymbol{\theta}) \in O(n^{-x})$, which implies that edges f which do not maximize $|f \cap e|$ will have negligible contributions to the likelihood. The essence of the argument is that forming x extant nodes requires x independent samples from the sets $\mathcal{V}_0 \setminus f$ and $\mathcal{V}_1 \setminus f$, each of which have size scaling linearly with n . So, each independent sample has probability $O(n^{-1})$, and the probability of forming a given set of x extant nodes is $O(n^{-x})$. We detail this argument in Section C.

In our application of simulated annealing, we run the algorithm for $A \times n$ steps where A represents the number of epochs. In each step, we propose a change to the labels $\mathbf{z}$ by generating a node index j uniformly at random and evaluating the change in likelihood of "flipping" the binary node label ($\mathbf{z}_j = 1 - \mathbf{z}_j$). If the proposal increases Equation (C9), it is accepted. However, if the proposal decreases the likelihood, we accept it with a probability that decreases as the epochs increase. When evaluating this change in likelihood, we implement custom data structures that allow us to compute only the change in likelihood from the prior state by storing the components of the prior likelihood calculation, resulting in a 30 times decrease in compute time on real world datasets at the expense of memory.

In simulated annealing applications, it is important accept proposals that decrease the value of the objective function with a probability that decreases gradually as the algorithm runs. Similarly, proposals that slightly lower the objective function should be accepted more frequently than proposals causing large objective function decreases. A balance must be struck, both accepting enough proposals decreasing the objective function to avoid becoming stuck in local optima and not accepting too many to allow convergence to a maxima. However, the change in log-likelihood of a proposal in our application changes significantly depending on the size of the hypergraph considered. Thus, traditional annealing schedules struggle to accept an effective balance of proposals that decrease the likelihood on varying sized hypergraphs. For example, on datasets with large m , the logged form of Equation (C9) have orders of magnitude larger changes compared to hypergraphs with small m . Thus, we implement a custom annealing schedule based on the idea of z-scores. Breaking the runtime into epochs, the algorithm's first epoch accepts all proposals while storing the changes in log-likelihood and, at the end

of the first epoch, calculating its standard deviation, $\sigma_{\mathcal{H}}$. For the rest of the steps of the algorithm, the acceptance probability of a proposal that decreases the likelihood is determined by the change in log-likelihood of the proposal divided by $\sigma_{\mathcal{H}}$, interpreting the change in likelihood in context of the specific hypergraph. A full specification of the algorithm is provided in Section D.

IV. RESULTS

How did we do on real and synthetic data?

We evaluate our methods on several data sets.

Contact high school: [1, 2] Contact primary school: [2, 36, 37] Senate bills: [2, 38, 39] Coauthorship: [40]

A. Parameter Estimation

We first illustrate the ability of SEM to recover true parameters from synthetic data in Figure 3. In the left-hand panel, show simulation results from [NUM REPEATITIONS]

Each synthetic hypergraph is generated using our model (VR: should the model have a name? PC: Yes, an ideal name will have an acronym which is fun in some way.) and a known parameter vector $\boldsymbol{\theta}$. We performed SEM on a set of hypergraphs which were generated with varying amounts of homophily across each of the three growth mechanisms: the edge copy step, the external node addition step, and the novel node addition step. We also varied the expected number of nodes added to e in each of these steps. To measure the success of a parameter recovery, we compute the Kullback-Leibler (KL) divergence between the true and estimated distributions described by each of the six parameters, and add the results. Using standard results, our measure of error then becomes

$$\begin{aligned} D[p_{\boldsymbol{\theta}}||\hat{p}_{\boldsymbol{\theta}}] = & \sum_{* \in \{+, -\}} c_* \left(\rho_* \log \frac{\rho_*}{\hat{\rho}_*} + (1 - \rho_*) \log \frac{1 - \rho_*}{1 - \hat{\rho}_*} \right) \\ & + \sum_{* \in \{+, -\}} \left(\gamma_* \log \frac{\gamma_*}{\hat{\gamma}_*} - \gamma_* + \hat{\gamma}_* \right) \\ & + \sum_{* \in \{+, -\}} \left(\eta_* \log \frac{\eta_*}{\hat{\eta}_*} - \eta_* + \hat{\eta}_* \right), \end{aligned}$$

where c is a factor describing, in expectation, how many nodes of each label are selected in the copy step. For the purposes of this analysis, we set $c_* = 1$ for both labels.

As seen in Figure 3, the quality of the learned parameter estimates depends on the seeded parameter vector. We observe especially low estimation errors when the seeded parameters reflect high copy rates ρ_* and homophily as reflected in the differences $\gamma_+ - \gamma_-$ and $\eta_+ - \eta_-$; a typical SEM run for these parameter settings is shown in the top row on the right of Figure 3. That

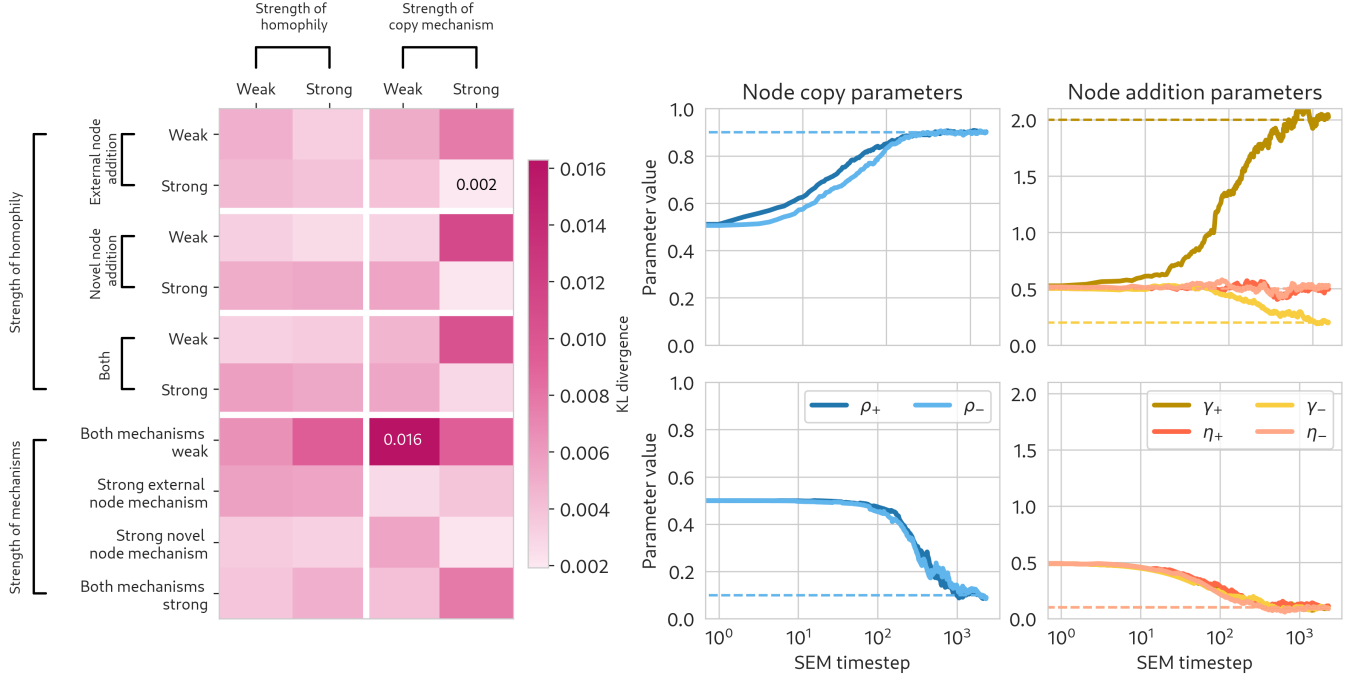


FIG. 3. Performance of SEM on synthetic hypergraphs. Each entry in the heatmap corresponds to a set of parameters θ , which can be found in Appendix G. For each set of parameters, five independent hypergraphs are generated according to the model and SEM is performed independently on each. **PC:** Violet, can we feasibly increase this to roughly 20 independent repetitions? We should also say how many simulation steps were used to generate each hypergraph. The Kullback-Leibler (KL) divergence between the true and estimated parameters is then computed for each of the five estimated parameter vectors $\hat{\theta}_i$ generated by SEM. The color of an entry in the heatmap represents the mean KL divergence over the five simulations for that set of parameters. The parameter estimates over iterations of SEM are plotted for the underlying parameters which produced the lowest mean KL divergence (top row) and the parameters that produced the highest mean divergence (bottom row).

TABLE I. Data sets used for parameter estimation. We show the number of nodes (n), edges (m), average degree ($\langle d \rangle$), and average edge size ($\langle k \rangle$) for each dataset.

	n	m	$\langle d \rangle$	$\langle k \rangle$	Nodes are	Edges are	Labels represent
senate-bills [8, 38, 39]	294	29157	789.6	8.0	Senators	Cosponsored bills	Political party
house-bills [8, 38, 39]	1494	60987	835.8	20.5	Congresspeople	Cosponsored bills	Political party
high-school [36, 37]	320	60544	484.6	2.6	Students	Social interaction	Gender
primary-school [1]	227	44626	590.7	3.0	Students	Social interactions	Gender
coauthorship [40]	105256	142834	3.4	2.5	Authors	Papers	Gender
emails VR: not sure what to cite here	14901	83372	18.5	3.3	Email addresses	Emails (sender and recipients)	Fringe vs. core

said, even in the least successful cases for estimation with high KL divergence, the learned estimates are still visually close to the true values, illustrated in the bottom row on the right of Figure 3.

We also used our method to infer parameters for six standard empirical hypergraph datasets described in Table I.

Figure 4 shows our parameter estimates for each of the empirical data sets. In each panel, we plot the same-label and opposite-label parameter estimates on opposing axes. Some data sets, like the gendered primary school contact network, appear on or near the line of equality in all three panels, suggesting that there is little selection in network evolution with respect to the chosen binary labels. The coauthorship data, meanwhile, falls on the line of equality for the edge copy and external node ad-

dition mechanisms, but there is a difference in the novel node addition parameters: $\hat{\eta}_+ > \hat{\eta}_-$. Our model presents a simplified picture of the process of collaboration formation as being initiated by a focal individual scientist, who then recruits from a previous collaboration, from other published scientists in the field, and from an unobserved pool of novices who have never published before. Our findings suggest that of these mechanisms, only the latter (the recruitment of novices) displays substantial gender homophily, with novices who share the gender of the focal scientist more likely to be recruited to the collaboration. This effect may be partially due to gender homophily between PhD students and their advisors, a known phenomenon in several scientific areas [41].

VR: begin NEW: discuss congressional data SEM

The senate-bills data appears to have slight homophily in the edge copy and external node addition steps, and has near-zero values for the novel node addition parameters. The house-bills data also has near-zero valued novel node addition parameters and has similar edge copy parameters to senate-bills (though house-bills is slightly heterophilic and senate-bills is slightly homophilic). The main distinction between the parameters of the two congressional networks is the external node addition step, where the values of both γ_+ and γ_- are higher in house-

bills than in senate-bills. This might be explained by the higher mean edge size of house-bills (20.5, as opposed to the 8.0 of senate-bills.). That is, a bill in the House of Representatives has on average more cosponsors than a bill in the Senate, and since only about half of f is copied to e , and η_- and η_+ are close to zero, the external node parameters will grow proportionally to the mean edge size.

VR: end NEW

The emails hypergraph falls in the heterophilic region of the edge copy panel. This is a feature of the sampling strategy. The enron dataset contains the email data from 145 users associated with the Enron Corporation. These users are the “core” nodes of the hypergraph. Any users who were included on emails with core users, but whose complete email data was not sampled, are referred to as “fringe” nodes. Each hyperedge represents an email between a set of nodes, or users. Therefore, each hyperedge must contain at least one core node. And since 73% of edges in the hypergraph have size 2, and less than one percent of the nodes in the hypergraph are core nodes, it is likely that any given edge contains one core node and one fringe node. In fact, 67% of edges in the Enron data match exactly this description. The high magnitude of the copy parameters, particularly ρ_- , is a result of the nature of email — rarely does an email interaction between a group of users occur only once. Often, an email is precipitated by an existing conversation between that set of users, making the edge copy mechanism well suited for modeling email network growth.

For all runs of SEM, the initial learning rate was $\alpha^{(0)} = 0.01$, which exponentially decayed over iterations according to the exponent 0.001. Initial sufficient statistics $\mathbf{s}^{(0)} = (1, 2, 1, 2, 0.5, 0.5, 0.5, 0.5)$ were used.

B. Simulated Annealing for Community Detection

We evaluate our simulated annealing algorithm on a range of small synthetic data sets under two conditions, shown in ???. Under the Nishimori condition [42], we generate a synthetic hypergraph according to a parameter vector θ and then attempt to learn $\mathbf{z}$ via simulated annealing using the same parameter vector θ . Under the “fixed” condition, which simulates a more realistic data analysis scenario, we always use the same parameter vector θ' to learn $\mathbf{z}$ via simulated annealing, regardless of the true parameter vector θ used to generate the synthetic data. In this condition, we use $\theta' = (0.9, 0.1, 0.2, 0.1, 2.0, 0.3)$, which is the same parameter vector used in Figure 2 and which we found to be a reasonable default for learning $\mathbf{z}$ across a range of synthetic data sets generated with different parameters. We compare these methods against each other and against a baseline of Laplacian spectral clustering applied to the weighted clique projection of the hypergraph.

Comparing the first and second rows of ??, we find that in these examples the penalty for using the “wrong”

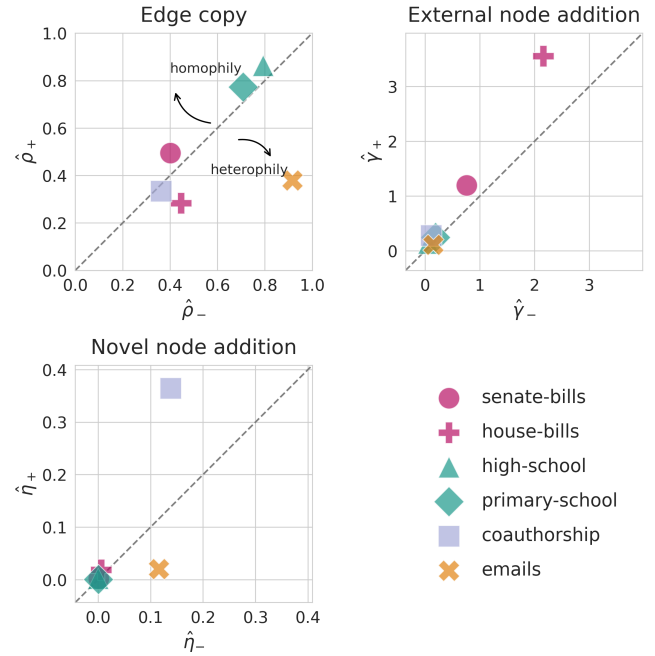


FIG. 4. Model parameters inferred via SEM for six empirical higher-order networks. **PC: imo we need text and points to be a bit bigger here for legibility, not sure if we can do this as a single-column**

parameter vector θ' is relatively small. Furthermore, maximum-likelihood estimation under both the Nishimori and fixed conditions tends to outperform clique-projection spectral clustering in most of the areas of parameter space explored by this experiment.

Further, we evaluate our simulated annealing algorithm on several empirical datasets: senate bills [2, 38, 39], contact high school [1, 2], and contact primary school [2, 36, 37]. To best simulate a real application where the parameters that formed the dataset are unknown, we use the same fixed parameter vector for all empirical datasets: $\theta' = (.9, .1, 1, .25, .001, .001)$. All of these datasets have small η under our model due to their large number of edges compared to small number of nodes. Further, especially in the context of the datasets, the novel node mechanism does not seem to be accurate for the treatment of nodes as the system and many of their nodes already exist far before the data is collected. Thus, in our empirical runs, we treat all nodes as external nodes in our simulated annealing algorithm.

FC: Ref greedy-modularity and spectral clustering FC: Ref ARI? FC: Talk about how the datasets were parsed? No duplicated hyperedges for the same interaction

In Figure 5, we observe that simulated annealing outperforms both greedy-modularity and spectral clustering on senate bills in all 10 runs in ARI. On the primary school dataset, simulated annealing performs comparably in ARI to the competitor methods, with a slightly lower average ARI compared to spectral clustering and greedy modularity’s results. However, simulated anneal-

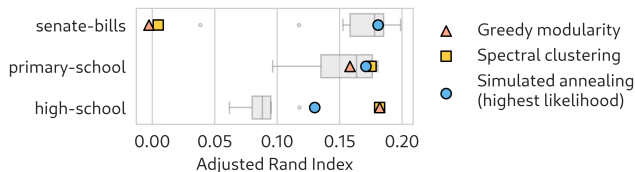


FIG. 5. Community detection results on empirical datasets. Box plots correspond to the max likelihood label set’s corresponding ARI plotted for 10 runs of simulated annealing on each dataset. We compare these results with the ARI of the labels generated from the spectral clustering and greedy modularity algorithms of community detection.

ing performs worse on the high school dataset in ARI compared to its competitors. While the computational performance of simulated annealing presents a difficulty in its application, the detection of communities via the maximization of likelihood provides evidence that the formations of these datasets, especially senate-bills, can be modeled via our proposed method.

FC: Unsure of the assumptions behind spectral clustering, tried to google but couldn’t find. If assumption of edge independence, could say something about that

V. DISCUSSION

We have proposed a simple model of binary-labeled hypergraph formation in which edges are formed through noisy copies of previous edges, combined with the addition of extant and novel nodes. Our model incorporates label homophily through each of the three edge formation mechanisms.

PC: More discussion

There are several directions for future work. Most obviously, although our results show that maximum-likelihood inference is a reasonable approach for com-

munity detection in this model, the simulated annealing method we have used is impractically expensive on data sets of even relatively modest size. Developing more efficient heuristics for maximizing the likelihood over $\mathbf{z}$ will be an important direction for scalable application of our proposed model.

There are also opportunities to extend the model to more realistic settings and more diverse data sets. One obvious extension is to nonbinary node-labels, which would enable multiway community detection algorithms. Another approach is to relax the assumption in our model that *every* edge is formed through the edge-copying mechanism; one could instead allow some edges to form spontaneously without copying an earlier edge. A final obvious extension is to incorporate recency-bias [23] into the copy mechanism, so that more recent edges are more likely to be copied while edges which are sufficiently old may cease to be copied at all.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

The code used to generate the results in this paper is available at <https://github.com/Violet-Ross/labeled-growth>. The empirical data sets analyzed in this paper are available at the following links:

- Contact high school:
- Contact primary school:
- Senate bills:
- Coauthorship:

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Appendix A: Power-Law Degree Distribution

Our derivation follows that of He et al. [25], which in turn is based on a derivation by Mitzenmacher [31].

At stationarity, the mean degree $\langle d \rangle$ of a node satisfies

$$\langle d \rangle = \frac{m}{n} \langle k \rangle, \quad (\text{A1})$$

where $\langle k \rangle$ is the mean edge size (computable from the stationary distribution $p_{K_0, K_1}(k_1, k_2)$), m is the number of edges, and n is the number of nodes. In expectation, there are $\eta \triangleq \eta_- + \eta_+$ new nodes added per new edge, which means that $\frac{m}{n} \rightarrow \frac{1}{\eta}$ as $m, n \rightarrow \infty$ in expectation. So, we have

$$\langle d \rangle = \frac{\langle k \rangle}{\eta}. \quad (\text{A2})$$

To proceed, we make a mean-field assumption that the degree d of a node is independent of the ratio $\frac{k_0}{k}$ of label-0 nodes in the edge f sampled to form the new edge e . Define the moments

$$\rho_{ij} = \left\langle \frac{k_i k_j}{k} \right\rangle. \quad (\text{A3})$$

We'll first calculate the expected number of nodes of label 0 sampled in the edge-sampling step. Suppose that an edge f is sampled with K_0 nodes of label 0 and K_1 nodes of label 1, and $K_0 + K_1 = K$ total nodes. Conditional on this event, the probability that a node of label 0 is selected as the seed node u is $\frac{K_0}{K}$, and when this occurs, the expected number of nodes of label 0 in e is $1 + \rho_+(K_0 - 1)$. If, on the other hand, a node of label 1 is selected as the seed node u , which occurs with probability $\frac{K_1}{K}$, then the expected number of nodes of label 0 in e is $\rho_- K_0$. So, the expected number of nodes of label 0 copied into the new edge e from f is

$$\sigma_0 \triangleq \left\langle \frac{k_0}{k} (1 + \rho_+(k_0 - 1)) + \frac{k_1}{k} \rho_- k_0 \right\rangle \quad (\text{A4})$$

$$= 1 + \rho_+ [\rho_{00} + \rho_{11} - 1] + \rho_- \rho_{01}. \quad (\text{A5})$$

Similarly, the expected number of nodes of label 1 copied into the new edge e from f is

$$\sigma_1 \triangleq \left\langle \frac{k_1}{k} (1 + \rho_+(k_1 - 1)) + \frac{k_0}{k} \rho_- k_1 \right\rangle \quad (\text{A6})$$

$$= 1 + \rho_+ [\rho_{00} + \rho_{11} - 1] + \rho_- \rho_{01}. \quad (\text{A7})$$

We let

$$\sigma \triangleq \sigma_0 + \sigma_1 = 1 + \rho_+(\rho_{00} + \rho_{11} - 1) + 2\rho_-\rho_{01}. \quad (\text{A8})$$

Importantly, σ depends only on the parameters and the moments of the joint distribution $p_{K_0, K_1}(k_1, k_2)$, which can be computed via linear algebra as described in Section II A 1.

We now invoke the mean-field assumption. Under this assumption, the probability that a single node selected in the sampling step has degree d is $dp_d^{(t)}$, where $p_d^{(t)}$ is the fraction of nodes with degree d at time t , since that node participates in d edges, independently of the ratio $\frac{k_0}{k}$ of label-0 nodes in the edge f sampled to form the

new edge e . The expected number of nodes of degree d sampled in the edge-sampling step is then $\sigma \frac{dp_d^{(t)}}{\langle d \rangle}$, and the expected *change* in the count of degree d nodes through the edge-sampling step is

$$\frac{\sigma}{\langle d \rangle} ((d-1)p_{d-1}^{(t)} - dp_d^{(t)}) \quad (\text{A9})$$

Similarly, the expected number of nodes which are sampled in the extant node addition step is $\gamma \triangleq \gamma_+ + \gamma_-$. Since this step does not involve any selection proportional to degree, the expected change in the count of degree d nodes through this step is

$$n^{(t+1)}p_d^{(t+1)} - n^{(t)}p_d^{(t)} = \frac{\sigma}{\langle d \rangle} [(d-1)p_{d-1}^{(t)} - dp_d^{(t)}] + \gamma(p_{d-1}^{(t)} - p_d^{(t)}). \quad (\text{A10})$$

We know that at stationarity, $n^{(t+1)} - n^{(t)}$ is in expectation equal to $\eta \triangleq \eta_+ + \eta_-$, the expected number of novel nodes, while at stationarity $p_d^{(t+1)} = p_d^{(t)} = p_d$ independent of t . Thus, we obtain

$$p_d \eta = \frac{\sigma}{\langle d \rangle} [(d-1)p_{d-1} - dp_d] + \gamma(p_{d-1} - p_d).$$

Rearranging, we obtain

$$\begin{aligned} \frac{p_d}{p_{d-1}} &= \frac{\frac{\sigma}{\langle d \rangle}(d-1) + \gamma}{\frac{\sigma}{\langle d \rangle}d + \gamma} \\ &= 1 - \frac{\frac{\sigma}{\langle d \rangle} + \eta}{\frac{\sigma}{\langle d \rangle}d + \gamma}. \end{aligned}$$

Allowing d to become large (which describes the tail of the degree distribution), we have

$$\begin{aligned} \frac{p_d}{p_{d-1}} &\approx 1 - \frac{1}{d} \frac{\frac{\sigma}{\langle d \rangle} + \eta}{\frac{\sigma}{\langle d \rangle}} \\ &= 1 - \frac{1}{d} \left(1 + \frac{\eta}{\sigma/\langle d \rangle} \right) \\ &= 1 - \frac{1}{d} \left(1 + \frac{\langle k \rangle}{\sigma} \right). \end{aligned}$$

a relation which corresponds to a power law with exponent $\zeta = 1 + \frac{\langle k \rangle}{\sigma}$. Explicitly calculating this exponent requires the moments $\langle k \rangle$, ρ_{00} , ρ_{11} , and ρ_{01} , which can be computed from the stationary joint distribution $p_{K_0, K_1}(k_1, k_2)$.

Appendix B: Sufficient Statistics for Stochastic Expectation-Maximization

We describe the function σ from Section III A which computes the conditional sufficient statistics for the pa-

rameters $\theta = (\rho_+, \rho_-, \gamma_+, \gamma_-, \eta_+, \eta_-)$ given an observed edge e , a candidate parent edge f , a candidate seed node u , and the node labels $\mathbf{z}$. Let $z = z_u$ be the label of the seed node u . Let e_z be the set of nodes in e with label z , and let f_z be the set of nodes in f with label z . Then,

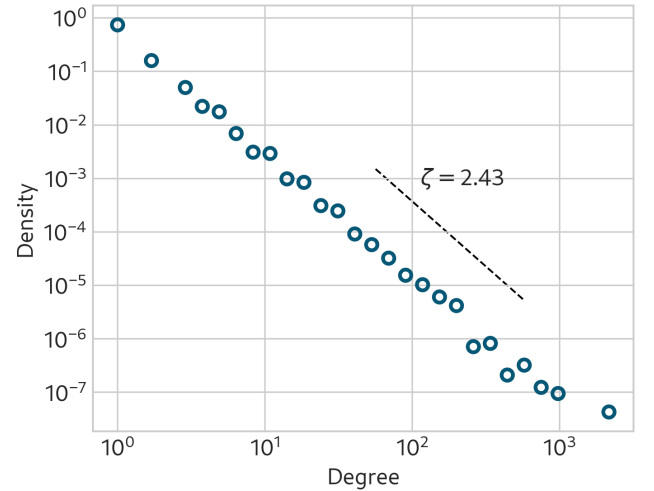


FIG. 6. Degree distribution from model simulation of 10^4 edges (points) and theoretical power law (line) with exponent given by Equation (2). The parameters in this experiment are $\rho_+ = 0.7, \rho_- = 0.3, \gamma_+ = 0.5, \gamma_- = \eta_-3, \eta_+ = 0.2, \eta_- = 0.1$.

the sufficient statistics for the parameters θ are given by

$$\sigma(e, f, u, \mathbf{z}) = \begin{pmatrix} |e_z \cap f_z| \\ |f_z| \\ |e_{\bar{z}} \cap f_{\bar{z}}| \\ |f_{\bar{z}}| \\ |(e_z \setminus f_z) \cap \mathcal{V}^{(t)}| \\ |(e_{\bar{z}} \setminus f_{\bar{z}}) \cap \mathcal{V}^{(t)}| \\ |e_z \setminus \mathcal{V}^{(t)}| \\ |e_{\bar{z}} \setminus \mathcal{V}^{(t)}| \end{pmatrix}^\top.$$

As described in Section III A, the vector $\mathbf{s}$ of expected sufficient statistics is computed as a weighted average of $\sigma(e, f, u, \mathbf{z})$ over the belief distribution $p(f, u | e, \hat{\theta}^{(t)})$ over the latent variables f and u . Then, the function g which computes the maximum likelihood estimates of the parameters θ from $\mathbf{s}$ is given by

$$g(\mathbf{s}) = \left(\frac{s_1 - 1}{s_2 - 1}, \frac{s_3}{s_4}, s_5, s_6, s_7, s_8 \right).$$

PC: Violet, I added the -1 here to what you had previously written but please check.

Appendix C: Approximation Argument for Simulated Annealing

This section provides further detail on the approximation to the likelihood described in Section III B and used in our simulated annealing implementation. The likelihood of an edge e being generated in timestep t can be written

$$\mathbb{P}(e | \mathbf{z}; \theta) = \sum_{\substack{f \prec_e \\ u \in f}} \mathbb{P}(e | f, u, \mathbf{z}; \theta) \quad (\text{C1})$$

$$= \frac{1}{m_t} \sum_{f \prec_e} \frac{1}{|f|} \sum_{u \in f \cap e} \mathbb{P}(e | f, u, \mathbf{z}; \theta), \quad (\text{C2})$$

where m_t is the number of edges at time t . It's helpful to reindex this sum by the size of the intersection $|f \cap e|$ between f and e .

$$\mathbb{P}(e | \mathbf{z}; \theta) = \sum_{h=0}^{|e|} \sum_{\substack{f \prec_e \\ |f \cap e|=h}} \frac{1}{m_t} \sum_{u \in f \cap e} \frac{1}{|f|} \mathbb{P}(e | f, u, \mathbf{z}; \theta). \quad (\text{C3})$$

We'll now argue that, when n is large, the terms in this sum corresponding to the largest intersections (for which $h = \arg\max_{f \prec_e} |f \cap e|$) will dominate the likelihood, allowing us to neglect edges f whose intersection with e is smaller.

Fix edge e and candidate parent edge f . We subscript e_z, f_z , and $\mathcal{V}_z$ to denote the subsets of e, f , and $\mathcal{V}$ with label z . Let $h_z = |(e_z \setminus f_z) \cap \mathcal{V}_z|$ be the number of extant nodes of label z in e . Then, the extant node addition

process appears in $\mathbb{P}(e | f, u, \mathbf{z}; \theta)$ as a factor

$$E(h_0, h_1) = \frac{\text{Poisson}(h_{z_u} | \gamma_+) \times \text{Poisson}(h_{\bar{z}_u} | \gamma_-)}{\binom{|\mathcal{V}_0^t| - |f_0|}{h_0} \binom{|\mathcal{V}_1^t| - |f_1|}{h_1}}, \quad (\text{C4})$$

where the numerator describes the likelihood of adding fixed numbers of extant nodes of each label and the denominator describes the likelihood of selecting the *specific set* of extant nodes from the set of all candidates. Under the assumption that $n \gg |f|, |e|$, we can estimate

$$\binom{|\mathcal{V}_z^t| - |f_z|}{h_z} \approx \frac{(|\mathcal{V}_z^t|)^{h_z}}{h_z!} \approx \frac{(c_z n)^{h_z}}{h_z!}, \quad (\text{C5})$$

where we have defined $c_z = \frac{|\mathcal{V}_z^t|}{n}$ as the fraction of nodes with label z . Then, absorbing terms that do not depend on n into a constant term $C(h_0, h_1)$, we have

$$E(h_0, h_1) = C(h_0, h_1) n^{-h_0-h_1} + O(n^{-h_0-h_1-1}) \quad (\text{C6})$$

$$= C(h_0, h_1) n^{-h} + O(n^{-h-1}), \quad (\text{C7})$$

where $h = h_0 + h_1 = |(e \setminus f) \cap \mathcal{V}|$ is the total number of extant nodes added to e and

$$C(h_0, h_1) = \frac{\text{Poisson}(h_{z_u} | \gamma_+) \times \text{Poisson}(h_{\bar{z}_u} | \gamma_-)}{\frac{(c_0)^{h_0}}{h_0!} \frac{(c_1)^{h_1}}{h_1!}}. \quad (\text{C8})$$

It follows that $\mathbb{P}(e | f, u, \mathbf{z}; \theta)$ is of order $O(n^{-h})$ as n grows large, where h is the number of extant nodes added to e . Terms corresponding to choices of f for which h is larger (of which there may be up to approximately m) decay quickly for large n . We therefore have

$$\mathbb{P}(e | \mathbf{z}; \theta) = \sum_{\substack{f \prec_e \\ |f \cap e|=h'}} \frac{1}{m_t} \sum_{u \in f \cap e} \frac{1}{|f|} \mathbb{P}(e | f, u, \mathbf{z}; \theta) + mO(n^{-h'-1}) \quad (\text{C9})$$

where $h' = \min_{f \prec_e} |(e \setminus f) \cap \mathcal{V}|$ is the minimum number of extant nodes added to e across all choices of f .

This argument suggests that, when n is relatively large in relation to m , the likelihood of e is dominated by terms corresponding to choices of f for which the number of extant nodes added to e is small. There are two limitations to this argument in the context of empirical data sets. First, it may be the case that only a small number of candidate parent edges f have small h , which may lead to a poor approximation of the likelihood when we evaluate only a small number of corresponding terms. Second, several of our data sets have m large in relation to n , making it dubious to neglect the error term. We therefore introduce two variations to our approximation heuristic:

- **Top j heuristic:** Rather than evaluate only terms for f for which h is minimal, we evaluate terms for the top j choices of f with the smallest h .

- **Batch normalization:** Rather than normalizing by $\frac{1}{m_t}$ in eq. (C9), we normalize by the number of f choices we evaluate for a specific e . This may be j under the top j heuristic, or it may be the number of f choices with h equal to the minimum h' if we do not use the top j heuristic. Letting $\mathcal{F}_e$ be the set of f choices we evaluate for a specific e , our approximation is

$$\tilde{\mathbb{P}}(e|\mathbf{z};\boldsymbol{\theta}) = \frac{1}{|\mathcal{F}_e|} \sum_{f \in \mathcal{F}_e} \sum_{u \in f \cap e} \frac{1}{|f|} \mathbb{P}(e|f, u, \mathbf{z}; \boldsymbol{\theta}). \quad (\text{C10})$$

Meaning the approximate likelihood of the entire $\mathcal{H}$ is

$$\tilde{\mathbb{P}}(\mathcal{H}, \mathbf{z}; \boldsymbol{\theta}) = \prod_{e \in \mathcal{E}} \tilde{\mathbb{P}}(e|\mathbf{z}; \boldsymbol{\theta}) \quad (\text{C11})$$

It is important to note that the batch normalization eq. (C11) tends to overestimate the likelihood, since we have restricted ourselves to an average of high-probability seed-edges. For the purposes of simulated annealing, this is not a major problem since we are only interested in the relative likelihood of different labelings $\mathbf{z}$ and not the absolute likelihood.

Appendix D: Simulated Annealing for Community Detection

We describe the simulated annealing algorithm implementation in detail. Define the number of epochs for the simulated annealing algorithm $A = 20$. Let B be the probability that the simulated annealing algorithm accepts a change on labels that lowers the approximated likelihood

Algorithm 2 SimulatedAnnealing($\mathcal{H}, \theta$)

Require: Hypergraph $\mathcal{H}$, parameters $\boldsymbol{\theta}$, number of epochs A
Ensure: Label set $\mathbf{z}$, approximation of the true labels $\mathbf{z}^*$
 Sample $\mathbf{z} \sim \text{Uniform}(\{0, 1\}^n)$
 $\Delta\mathcal{L} \leftarrow \emptyset$
 $\sigma_{\mathcal{H}} \leftarrow 0$
for $t = 0$ to A **do**
 for $\tau = 0$ to n **do**
 Sample $j \sim \text{Uniform}(\{0, \dots, n-1\})$
 $\mathbf{z}' \leftarrow \mathbf{z}$
 $z'_j \leftarrow 1 - z_j$
 $\Delta E \leftarrow \log \tilde{\mathbb{P}}(\mathcal{H}, \mathbf{z}'|\boldsymbol{\theta}) - \log \tilde{\mathbb{P}}(\mathcal{H}, \mathbf{z}|\boldsymbol{\theta})$
 if $t = 0$ **then**
 $\Delta\mathcal{L} \leftarrow \Delta\mathcal{L} \cup \{\Delta E\}$
 if $\sigma_{\mathcal{H}} = 0$ & $t > 0$ **then**
 $\sigma_{\mathcal{H}} \leftarrow \sigma(\Delta\mathcal{L})$
 if $\Delta E > 0$ **then**
 $\mathbf{z} \leftarrow \mathbf{z}'$
 else
 $B \leftarrow 1$
 $\zeta_a \leftarrow \Delta E / \sigma_{\mathcal{H}}$
 if $1 \leq t \leq 4$ **then**
 Sample $B \sim \text{Normal}(\zeta_a | \mu = 0, \sigma = 2)$
 else if $5 \leq t$ **then**
 Sample $B \sim \text{Normal}(\zeta_a | \mu = 0, \sigma = 2(1 - \frac{t}{A}))$
 Sample $x \sim \text{Uniform}(0, 1)$
 if $B > x$ **then**
 $\mathbf{z} \leftarrow \mathbf{z}'$
return $\mathbf{z}$

Here,

$$\sigma(\Delta\mathcal{L}) = \sqrt{\frac{\sum_{\delta \in \Delta\mathcal{L}} \delta^2 - (\frac{1}{|\Delta\mathcal{L}|} \sum_{\delta \in \Delta\mathcal{L}} \delta)^2}{|\Delta\mathcal{L}| - 1}} \quad (\text{D1})$$

Appendix E: Simulated Annealing Illustration

In this simulation, we generated a simple hypergraph according to our model with 61 nodes and 200 edges. Running simulated annealing for 1220 steps, we obtained a maximum log-likelihood of -9.49 nats per edge with an Adjusted Rand Index (ARI) against ground-truth of 0.18. Greedy modularity maximization as implemented in the Python package `networkx` [43] obtained an ARI of 0.04. The parameters for the model used to generate the hypergraph are

$$\begin{aligned} \rho_+ &= 0.90 \\ \rho_- &= 0.10 \\ \gamma_+ &= 1.00 \\ \gamma_- &= 0.25 \\ \eta_+ &= 0.20 \\ \eta_- &= 0.10. \end{aligned}$$

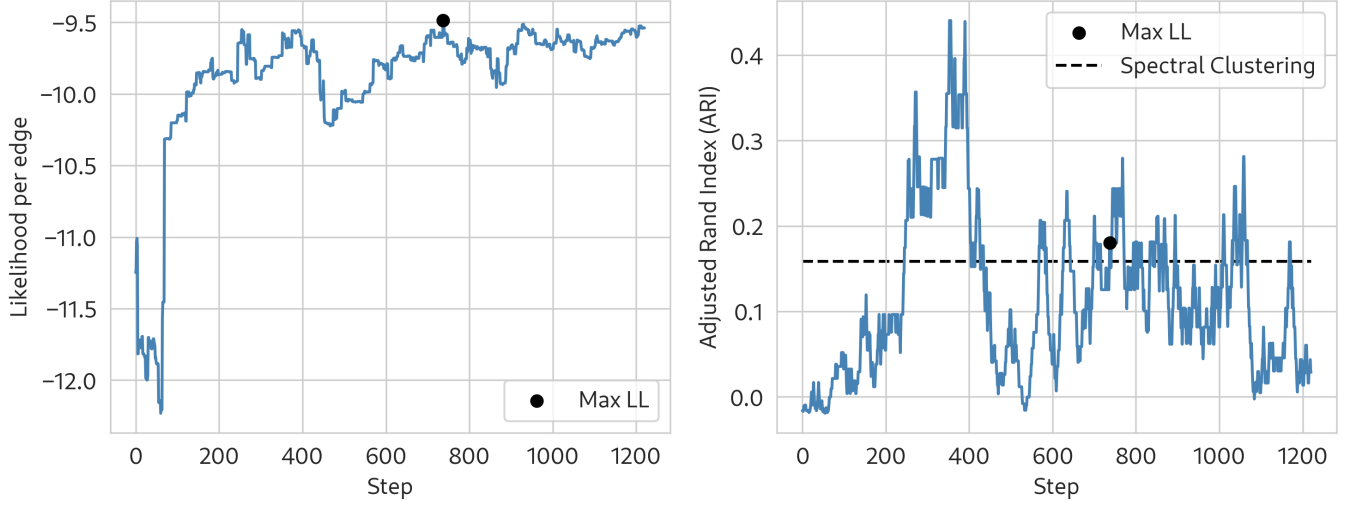


FIG. 7. Illustrative run of simulated annealing for community detection in a hypergraph generated from our model.

The guessed parameters used to compute the likelihood during simulated annealing are

$$\begin{aligned}
 \hat{\rho}_+ &= 0.90 \\
 \hat{\rho}_- &= 0.10 \\
 \hat{\gamma}_+ &= 1.00 \\
 \hat{\gamma}_- &= 0.25 \\
 \hat{\eta}_+ &= 0.20 \\
 \hat{\eta}_- &= 0.10 .
 \end{aligned}$$

TABLE III. Strength of Homophily in External and Novel Node Addition

		γ_+	γ_-	η_+	η_-
External node addition	Strong	1.0	0.8	0.2	0.2
	Weak	2.0	0.2	0.2	0.2
Novel node addition	Strong	0.2	0.2	1.0	0.8
	Weak	0.2	0.2	2.0	0.2
Both	Strong	1.0	0.8	1.0	0.8
	Weak	2.0	0.2	2.0	0.2

Appendix F: More Simulated Annealing on Synthetic Data

Appendix G: Synthetic SEM Parameter Values

TABLE II. Strength of Homophily in Edge Copying

	ρ_+	ρ_-
Weak	0.6	0.4
Strong	0.9	0.2

TABLE IV. Strength of Copy Mechanism

	ρ_+	ρ_-
Weak	0.1	0.1
Strong	0.9	0.9

TABLE V. Strength of External and Novel Node Addition Mechanisms

	γ_+	γ_-	η_+	η_-
Both weak	0.1	0.1	0.1	0.1
Strong external node mechanism	2.0	2.0	0.2	0.2
Weak novel node mechanism	0.2	0.2	2.0	2.0
Both strong	2.0	2.0	2.0	2.0